
INDUSTRY EXPERIENCE

- **AbbVie Inc.** North Chicago, USA
Senior Data Scientist Feb'23 - Present
 - **Anomaly Detection in Clinical Trials:** Led the design and deployment of clinical domain-agnostic GLMM-based unsupervised anomaly detection model to detect malpractices. Identified 3 fraudulent sites in 5 AbbVie trials, preventing further risk to upcoming studies and FDA dismissal.
 - Designed novel statistical testing framework with time similarity test and ANOVA variant to flag discrepancies in ePRO diaries. Enhanced data integrity and compliance across portfolio.
 - **Duplicate Patient Detection:** Spearheaded design and development of duplicate patient detection pipeline in Python, allowing pre-dosing interventions and reducing risk of statistical dilution of trial results - major advancement over current methods that rely solely on post-dosing data.
 - Performed Louvain community detection on geographical network of sites to reduce scope of comparisons by 95%.
 - Used PCA to create patient profiles from clinical data and vector similarity metrics to score subject pairs. Flagged duplicates with 80% accuracy by performing outlier detection on similarity scores.
 - Reduced pipeline runtime by 86% by parallelizing code and parsed 500+ studies in less than 30 mins.
 - **Risk Based Quality Management of Clinical Trials:** Created a risk KPI using occurrence of protocol deviations for portfolio-wide real-time monitoring and risk management.
 - **Immunology Trial Operations:** Categorized performance of sites based on enrollment trends using time series k-means. Reduced study startup expenditure by \$500k by highlighting sites with high enrollment potential.
- **Wyss Institute, Harvard University** Boston, USA
Research Associate Feb'19 - Aug'21
 - **Biostasis (DARPA) & Coronavirus Therapeutics (DARPA):** Built a computational drug discovery pipeline with Markov Random Field in Python; validated to be 300% faster throughput than traditional processes.
 - Performed graphical inference to identify drug targets for COVID-19, producing 3 hits leading to clinical trials.
 - Formulated a drug clustering pipeline with HDBSCAN and agglomerative clustering to streamline experimental testing of cluster representatives and chemical space exploration. Reduced experiment backlog by 1.5 weeks.
 - Led development of proprietary drug discovery Python package *NeMoCAD*, now licensed to *Unravel Biosciences*.
- **Vertex Pharmaceuticals** Boston, USA
Data Strategy and Solutions Intern May'22 - Aug'22
 - **International Drug Compliance:** Created a compliance scoring algorithm for European markets with drug supply volumes as input, providing visibility into adherence trends and enabling data-driven strategic decisions.

EDUCATION

- **Master of Science in Data Science** 2021-2022
Columbia University, Boston, USA
- **Master of Science in Mathematics** 2016-2018
Indian Institute of Technology Delhi, India

AWARDS & PUBLICATIONS

- **AbbVie 2024 R&D President's Award:** Innovative solution and accelerated delivery of anomaly detection tool.
- R. Novack, T. Lin, **S. Kaushal**, et al.(2025). **AI-enabled drug prediction and gene network analysis reveal therapeutic use of vorinostat for Rett Syndrome in preclinical models.** *Commun Med* 5, 249 (2025). Published in Nature.
- M. Sperry, T. Oskotsky, I. Maric, **S. Kaushal**, et al.(2023). **Target-agnostic drug prediction integrated with medical record analysis uncovers differential associations of statins with increased survival in COVID-19 patients.** *PLOS Computational Biology*, 19(5), e1011050.
- Co-authored *ACS Nano* (2024) paper on torpor-inducing drug predicted through Bayesian network tool (Biostasis).

PROJECTS

- **Johnson & Johnson Capstone:** Built a family of Random Forest models to predict commercial insurance payments for 40 different surgical procedures using IBM MarketScan CCAE database as input, outperforming J&J model by 10%.

SKILLS

Languages: Python, R, SQL | **Tools:** AWS, GIT | **Algos:** NLP, Neural Networks, Causal Inference, k-NN, EDA